Uncertainty Calibration of Multi-Label Bird Sound Classifiers

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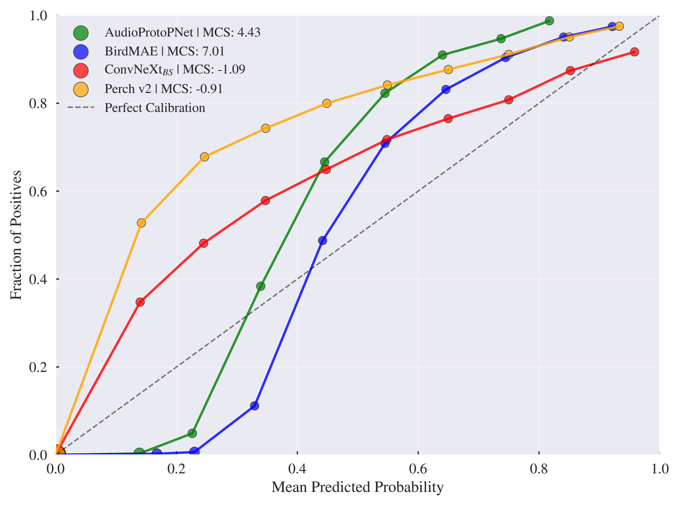
Introduction

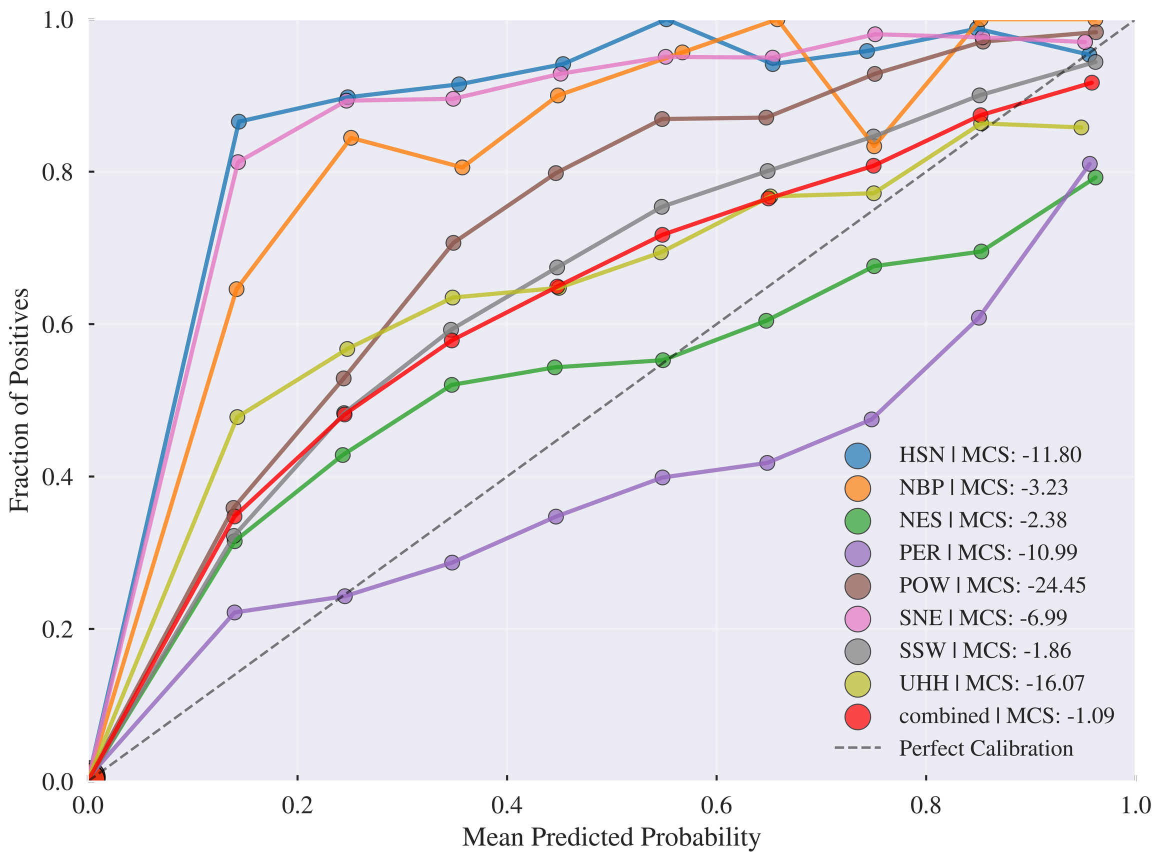
- Passive acoustic monitoring (PAM) enables large-scale biodiversity monitoring, but reliable species detection requires well-calibrated uncertainty estimates from deep learning classifiers
- Calibration is challenged by overlapping vocalisations, long-tailed species distributions, and distribution shifts between training and deployment data
- Overconfident models can bias occupancy predictions, while underconfident models reduce utility for expert review—reliable uncertainty is critical for transparent environmental decision-making
- The calibration of multi-label deep learning classifiers in bioacoustics has not yet been systematically assessed

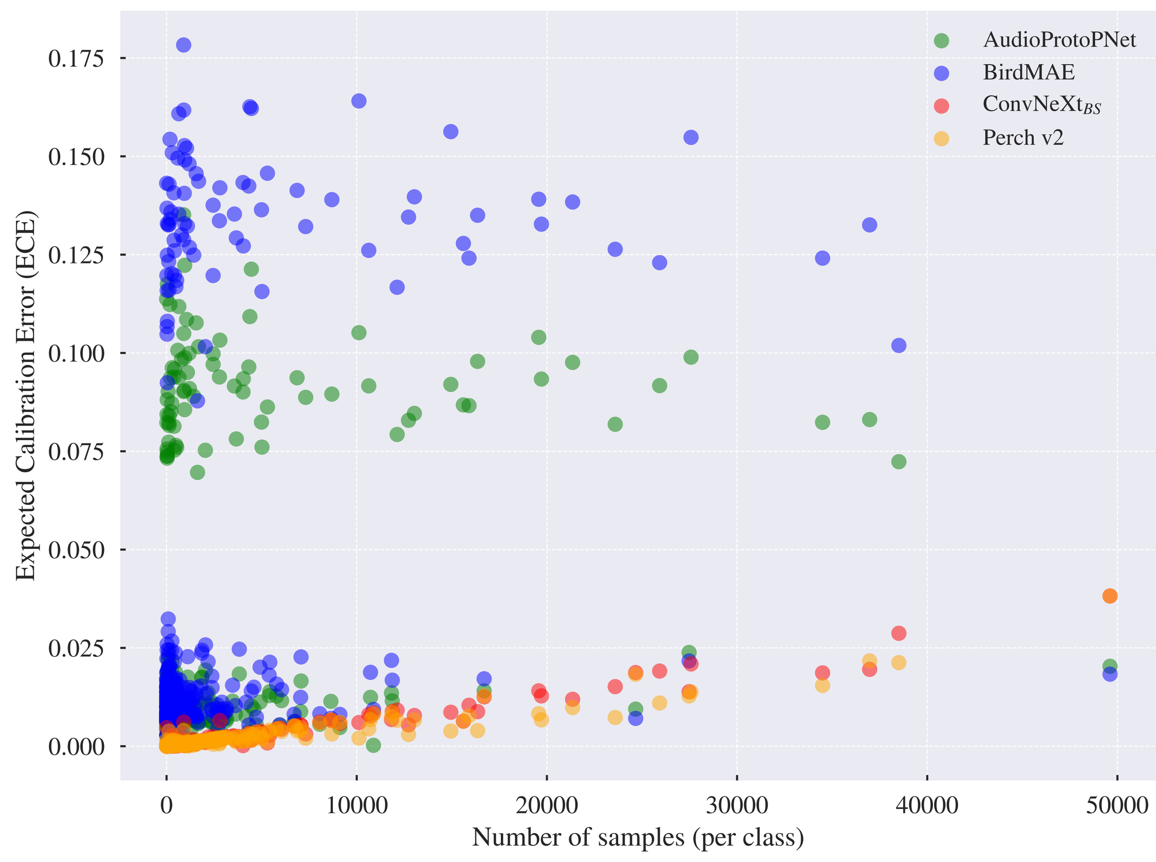
Contributions

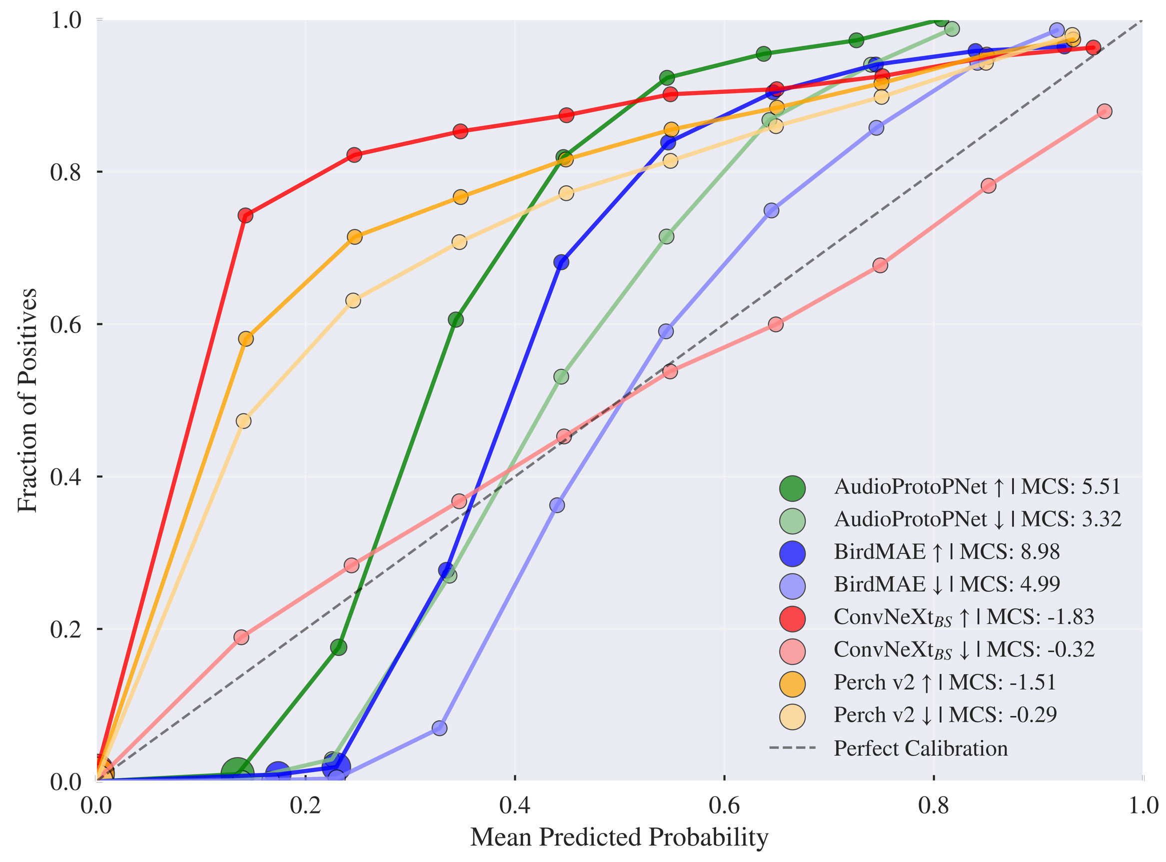


- **C1.** Benchmark calibration of four state-of-the-art multi-label bird sound classifiers on BirdSet, reporting global, per-dataset and per-class calibration with threshold-free metrics
- **C2.** Investigate post hoc strategies (temperature and Platt scaling): Platt scaling with 10 minutes of test data can significantly improve calibration
- **C3.** Summarise implications for practitioners: deployment-specific calibration evalu-

Benchmarking the Calibration of Bird Sound Classifiers







Experimental setting. Models: AudioProtoPNet, BirdMAE, ConvNeXt$_{\mathrm{BS}}$, Perch v2 (multi-label classifiers on BirdSet). **BirdSet:** Eight expert-annotated test datasets from PAM deployments. **Metrics:** ECE (expected calibration error), MCS (miscalibration score), OCS/UCS (over/underconfidence); cmAP for discrimination. **Results.** Globally, Perch v2 and ConvNeXt$_{\mathrm{BS}}$ are underconfident; AudioProtoPNet and BirdMAE are overconfident. Calibration varies significantly per dataset; combined ECE can mask domain-specific miscalibration, so per-dataset evaluation is essential. Class-wise ECE (centre figure) shows Perch v2 and ConvNeXt$_{\mathrm{BS}}$ with low, well-structured ECE; AudioProtoPNet and BirdMAE exhibit larger spread. For common vs. rare species (right), rare classes shift toward overconfidence for AudioProtoPNet and BirdMAE; Perch v2 and ConvNeXt$_{\mathrm{BS}}$ remain underconfident for both subsets. Class-wise calibration appears better for less-frequent classes.

Post-hoc Calibration

- Temperature scaling (TS) and Platt scaling (PS) evaluated with two calibration sets: global parameters on POW and per-class parameters on the first 10 minutes of each test dataset
- Per-class Platt scaling with 10 minutes of labelled data provides the most reliable calibration improvement
- Global scaling benefits Perch v2 and ConvNeXt$_{\mathrm{BS}}$ but can degrade AudioProtoPNet and BirdMAE on certain datasets
- No method consistently outperforms across all models and datasets—calibration improvement is domain-specific

Conclusions

- Calibration varies substantially across datasets and classes; aggregating across domains can mask systematic miscalibration
- Perch v2 and ConvNeXt$_{\mathrm{BS}}$ consistently underconfident; AudioProtoPNet and BirdMAE show mixed over-/underconfidence
- Simple post hoc methods (Platt scaling with 10 min labelled data) can significantly improve deployment-specific calibration
- Recommend reporting OCS/UCS alongside ECE or MCS to reveal whether errors stem from over- or underconfidence
- Future work: Bayesian approaches, calibration@k metrics, and evaluation across additional deployment scenarios

KEY REFERENCES

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